

Department of Computer Science & Engineering

Global Campus, Jakkasandra Post, Kanakapura Taluk, Ramanagara District, Pin Code: 562 112

Team No: 31

Hemo-Sclera+: A Multimodal AI-Powered Medical Diagnosis System

Presentation By,

A. VISWATEJA	(23BTRCL034),
UDHAYAN.G	(23BTRCL024),
V.MADHAVA SARMA	(23BTRCL239),
SRIDHAR.S	(23BTRCL080)
GNANENDRA REDDY .N	(23BTRCL226)
M. SURYA JASHWANTH	(23BTRCL214)

Guided By,

Dr. Sugumar R M .Tech, Ph.D.,
Associate Professor / CSE- AIML

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ABSTRACT

This project focuses on developing **Hemo-Sclera+**, an AI-powered, non-invasive medical diagnosis system to detect Anemia and Jaundice using computer vision techniques. The system analyzes eye regions such as the palpebral conjunctiva and sclera to identify subtle color variations associated with hemoglobin and bilirubin levels, supporting early detection. A structured deep learning pipeline was followed, including image preprocessing, segmentation using

A deep learning-based segmentation model (YOLOv11-Nano) was used as the baseline for accurate tissue detection, while a Random Forest regressor served as the primary predictive model due to its strong performance in estimating hemoglobin and bilirubin levels from extracted features. Additional approaches, including transformer-based neural networks, were implemented to improve contextual understanding and robustness. The final system demonstrates how artificial intelligence can assist healthcare professionals by providing faster, non-invasive, and reliable diagnostic insights for early screening.

INTRODUCTION

- AI-based system for Anemia and Jaundice detection
- Uses eye image data (palpebral conjunctiva and sclera) for analysis
- Focus on non-invasive early screening and triage support
- Deep learning models used for segmentation and feature extraction
- Improves accuracy, accessibility, and reliability in

Here's a breakdown of key introductory information:

1) **AI-Based Medical Diagnosis:**

The project focuses on detecting Anemia and Jaundice using deep learning models trained on eye image data (palpebral conjunctiva and sclera).

2) **Early Detection Focus:**

3) Data-Driven Approach:

Image data is processed through preprocessing, segmentation, and feature extraction (CIE $L^*a^*b^*$ color space) to identify meaningful diagnostic patterns.

4) Multiple Model Evaluation:

Models such as YOLO-based segmentation, Random Forest regression, and transformer-based architectures are used to improve accuracy and robustness.

SYSTEM REQUIREMENT

Software CONFIGURATION

- Operating system : Mac OS
sequoia/Windows OS
- Software : MATLAB R2024B

Hardware CONFIGURATION

- System : Intel i5/M1/M2
CHIP/I5
- Hard disk : 256 GB
- RAM : 8 GB

LITERATURE SURVEY

1. [World Health Organization](#). Global Surveillance, Prevention and Control of Anemia and Nutritional Disorders; Geneva, Switzerland, 2024.
2. [World Health Organization](#). Guidelines for the Management of Neonatal and Adult Jaundice; Geneva, Switzerland, 2023.
3. [International Society of Blood Transfusion](#). Global Status of Hemoglobin Disorders and Anemia Detection, 2025.

4. Rahaman, M.M.; Li, C.; et al. Deep Learning-Based Medical Image Analysis: Applications in Disease Detection; J. Medical Imaging & AI, 2024.
5. Yahiaoui, A.; Yumusak, N. Automated Disease Diagnosis Using Machine Learning Techniques; Biomedical Research, 2025.

EXISTING SYSTEM

The application of AI and computer vision in medical diagnosis is an emerging area within healthcare technology. Most existing systems focus on image-based or lab-based diagnostics.

Here's a breakdown of current systems and their characteristics:

Key Applications:

- **Clinical Decision Support Systems – –**

- **Medical Imaging Systems** – analyzing images like X-rays, CT scans, or retinal scans using deep learning
(e.g • Image Classification • Disease Detection)•
- Population Health Analytics – analyzing disease trends using patient data. (E.g • Surveillance
- Prevention)
- **Telemedicine & Symptom Checker Apps** – providing basic health assessment during

Existing Systems and Technologies:

- Machine Learning Algorithms:
- Common algorithms used include:

- YOLO-based Object Detection /

Segmentation

- Convolutional Neural Networks (CNN)
- Random Forest Regressor
- Transformer-based Models

- Examples of AI in use:
- AI models detect Anemia and Jaundice using eye image analysis
- Deep learning used for non-invasive screening and real-time prediction
- Analyzes color variations in **palpebral conjunctiva and sclera**
- Multiple models (YOLO, Random Forest, Transformers) used for better accuracy

Robustness:

The system must handle variations such as lighting, motion blur, and image quality for reliable real-world usage.

Ethical & Clinical Validation:

AI models should be clinically validated and

PROPOSED SYSTEM

System Overview

The AI-powered system detects [Anemia](#) and [Jaundice](#) using computer vision and deep learning techniques. The system analyzes eye images (palpebral conjunctiva and sclera) to identify color-based biomarkers related to hemoglobin and bilirubin levels. Image preprocessing, segmentation, and model-based prediction ensure accurate and reliable results. The system supports early detection, improves diagnostic efficiency, and acts as a non-invasive decision-support tool for healthcare professionals.

Key Components:

1. Data Collection Module:

- Collects eye images using smartphone or webcam
- Focuses on palpebral conjunctiva and sclera regions
- Uses annotated medical image datasets and clinical reference values (Hb & bilirubin)

2. Data Preprocessing Module:

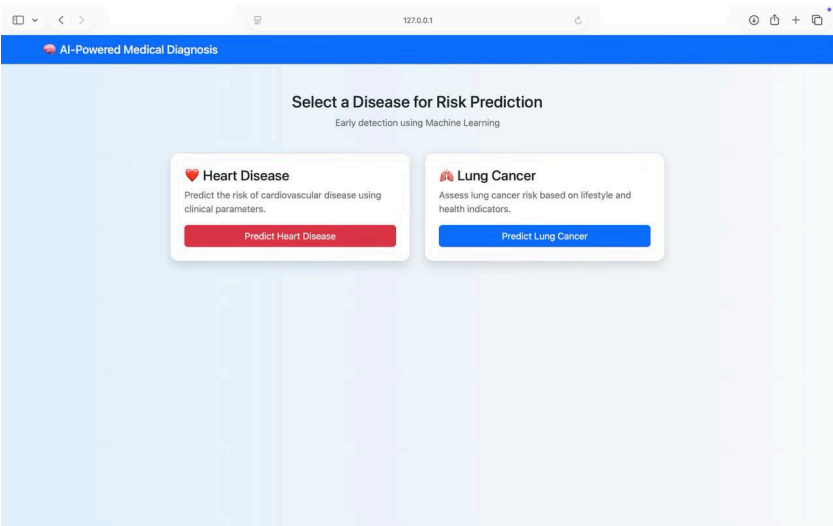
- Image resizing and normalization
- Noise reduction using Gaussian filtering
- Contrast enhancement using CLAHE
- Data augmentation (rotation, flipping, scaling) for robust model training

3. Feature Engineering Module:

- Extracts meaningful features from eye images in the CIE $L^*a^*b^*$ color space
- Identifies key biomarkers such as redness (a^*) for hemoglobin and yellowness (b^*) for bilirubin
- Captures texture and color distribution patterns from the conjunctiva and sclera

WORKING MODEL

Website Home page:



Heart Disease Risk analysis:

Heart Disease Risk Prediction

Enter Patient Health Details

Age (years)

Gender

Select

Diabetes

Yes

High Blood Pressure (Hypertension)

Yes

Obesity

Yes

Smoking Habit

Smoker

Alcohol Consumption

Consumes Alcohol

Physical Activity Level

Low

Diet Score (0 = Poor, 10 = Excellent)

Total Cholesterol (mg/dL)

Triglycerides (mg/dL)

LDL (Bad Cholesterol)

HDL (Good Cholesterol)

Systolic BP (Upper Value)

Diastolic BP (Lower Value)

Air Pollution Exposure

High

Family History of Heart Disease

Yes

Stress Level

Low

Access to Healthcare

Good

Previous Heart Attack

Yes

Emergency Response Time (minutes)

Annual Income (₹)

Health Insurance

Yes

Predict Heart Disease Risk

Lung Cancer Risk analysis:

Lung Cancer Risk Prediction

Enter Patient Respiratory & Lifestyle Details

Age (years)

Gender

Select

Smoking Habit

Smoker

Finger Discoloration (Yellow Fingers)

Present

Mental Stress

Low

Exposure to Pollution

High

Long-Term Illness

Yes

Energy Level (0 = Very Low, 1 = High)

Weak Immune System

Yes

Breathing Difficulty

Yes

Alcohol Consumption

Consumes Alcohol

Throat Discomfort / Chronic Cough

Present

Oxygen Saturation (%)

Chest Tightness / Pain

Yes

Family History of Lung Disease

Yes

Smoking History in Family

Yes

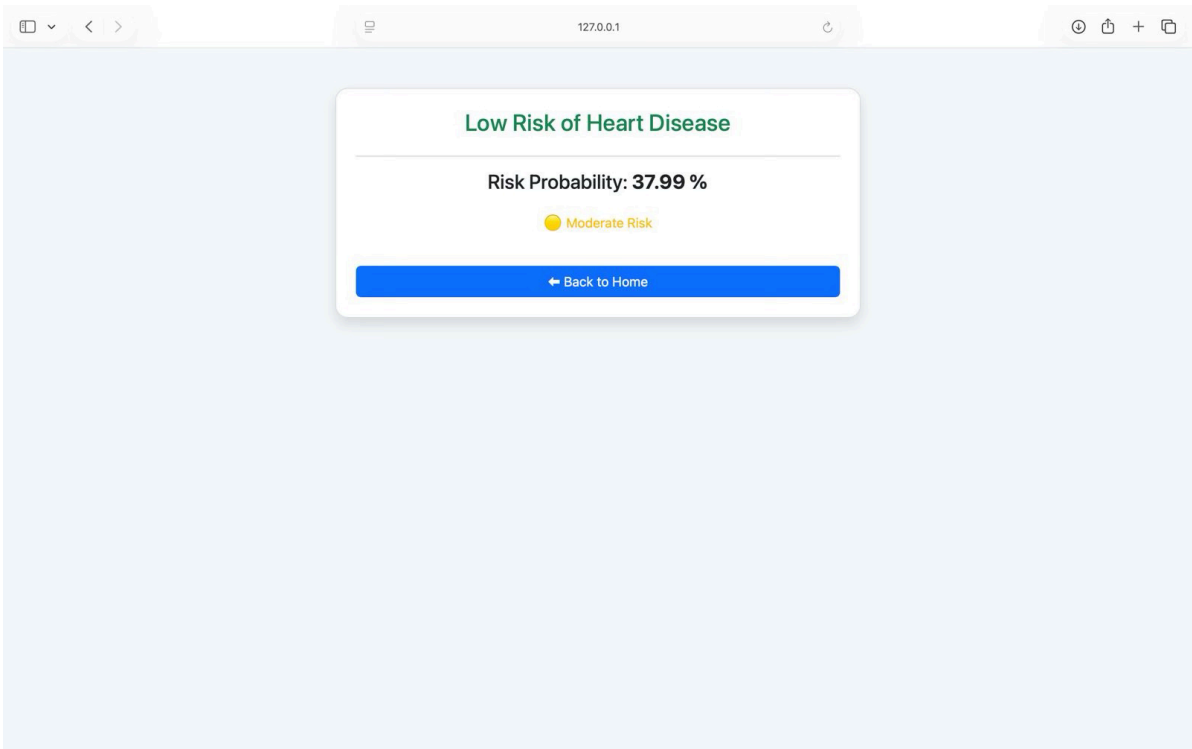
Stress Affecting Immunity

Yes

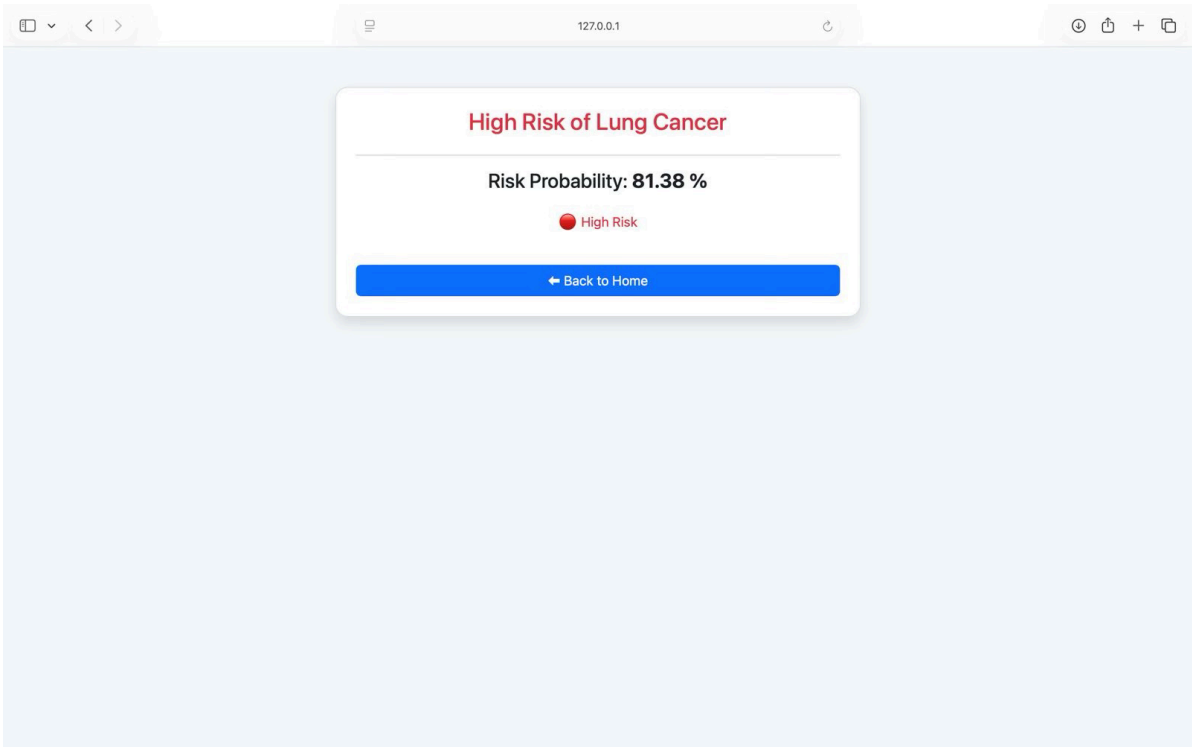
Predict Lung Cancer Risk

[Back to Home](#)

Heart Disease Output:



Lung Disease Output:



Conclusion:

- Hospitals and healthcare providers can use AI-based vision systems like Hemo-Sclera+ to detect Anemia by analyzing eye regions such as the palpebral conjunctiva for signs of pallor.
- Clinical decision-support tools can assist doctors in identifying Jaundice early by analyzing the yellow discoloration in the sclera using computer vision techniques.

Preventing healthcare disparities requires a

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THANK YOU